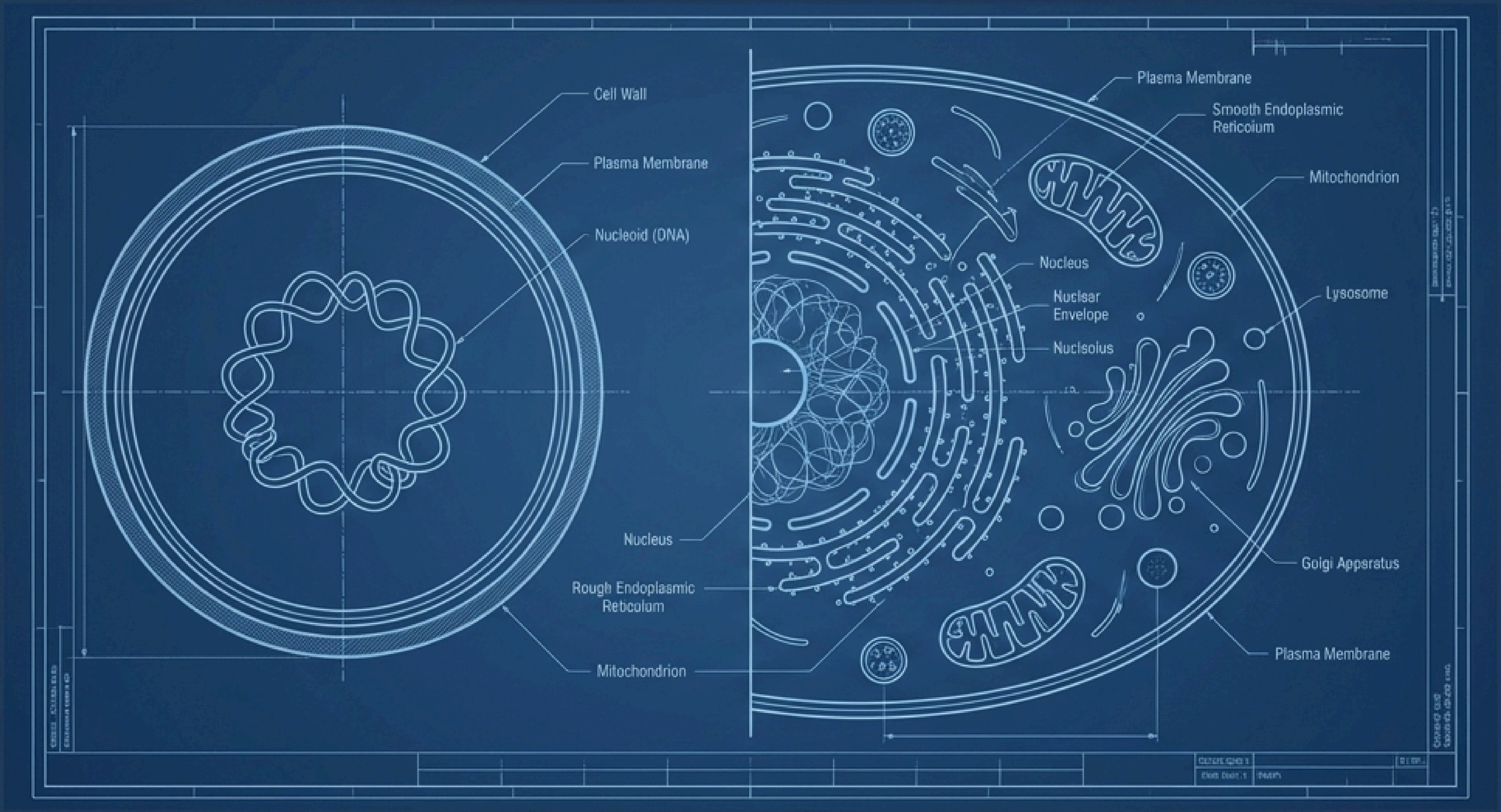
The Cell: An Architectural Blueprint of Life

A Detailed Exploration of Prokaryotic and Eukaryotic Structures for NEET-UG



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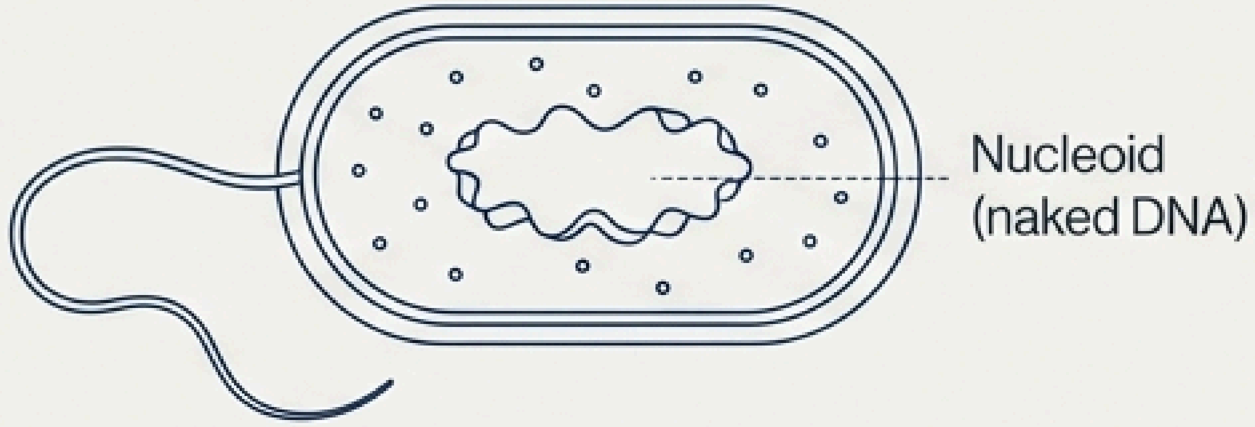
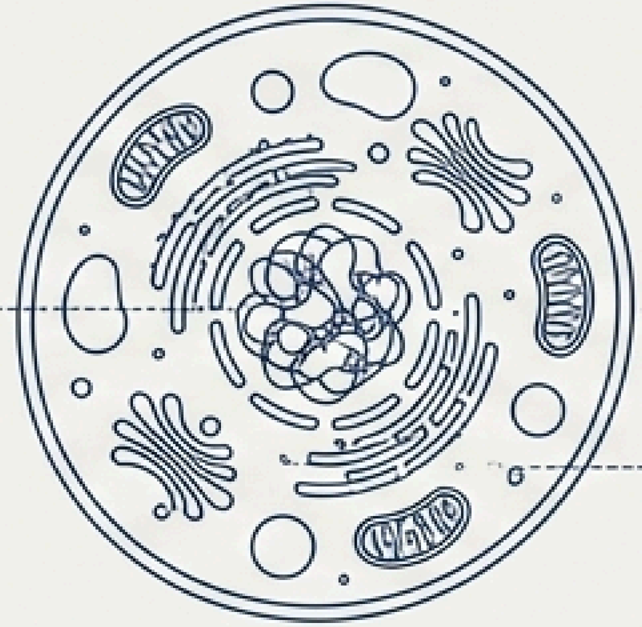
Two Master Plans: The Core Distinction





Feature	Prokaryotic Cell (The "Studio Apartment")	Eukaryotic Cell (The "Multi-Room Mansion")
Nucleus	Absent. Genetic material (nucleoid) is 'naked' in the cytoplasm.	Present. True nucleus with a nuclear envelope.
Organelles	No membrane-bound organelles.	Membrane-bound organelles present.
Size & Speed	Generally smaller and multiply more rapidly.	Generally larger and multiply more slowly.
Genetic Material	Single circular chromosome and often plasmids.	Multiple chromosomes.
Ribosomes	70S type.	80S in cytoplasm, 70S in organelles.

Blueprint of a Prokaryote: Simplicity and Variety

Prokaryotic Representatives

Bacteria, Blue-Green Algae, Mycoplasma, and PPLO (Pleuro Pneumonia Like Organisms).

Four Basic Shapes

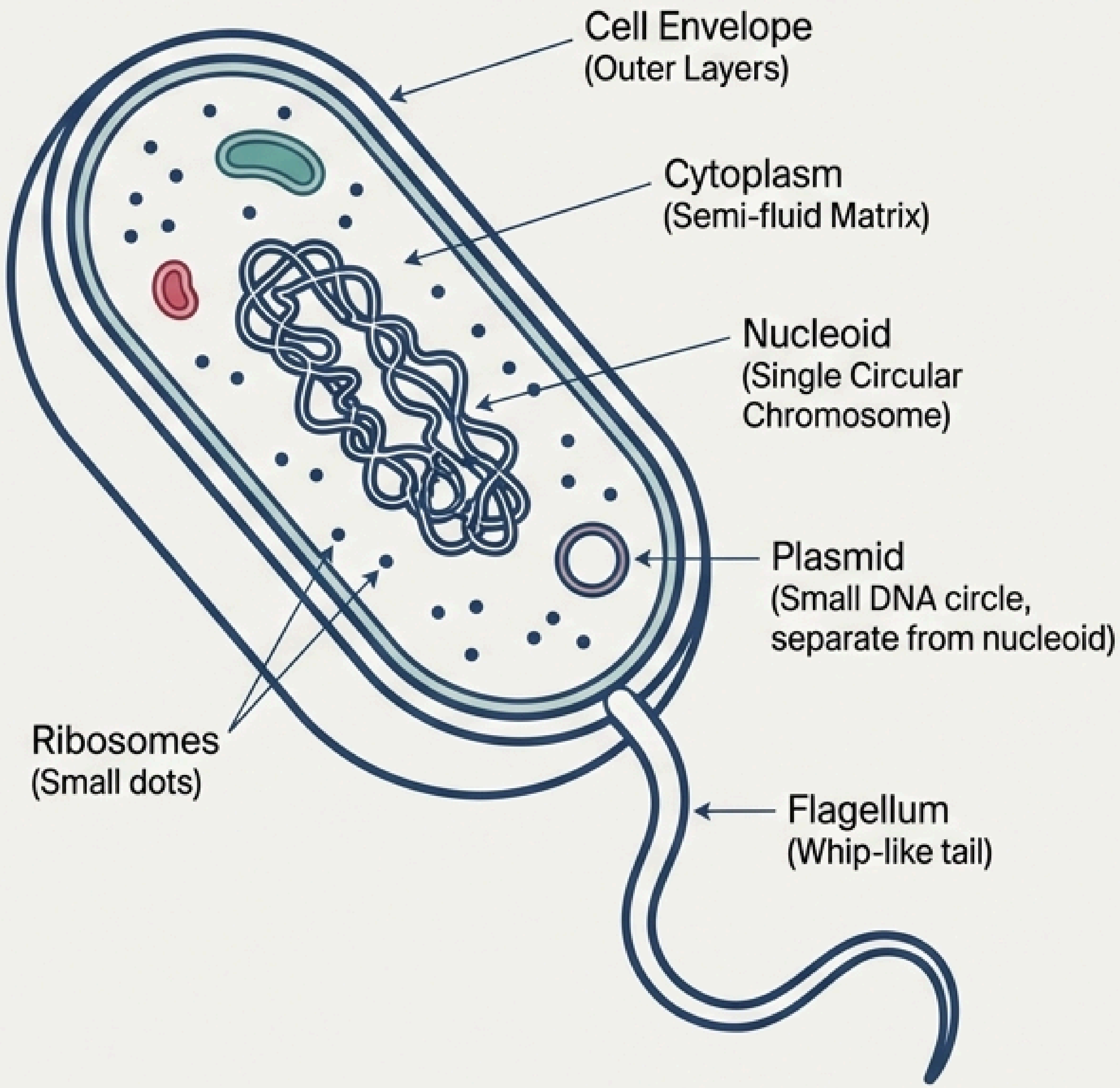
Bacillus (rod-like)	Coccus (spherical)	Vibrio (comma-shaped)	Spirillum (spiral)

Fundamental Organization

Despite varied shapes, the internal organization is fundamentally similar. The cell is filled with a semi-fluid matrix, the cytoplasm.

Unique Architectural Features

- The genetic material is naked, not enveloped by a nuclear membrane. It is a single, circular chromosome.
- Many bacteria have small, circular DNA called plasmids outside the genomic DNA, which can confer traits like antibiotic resistance.
- No membrane-bound organelles are found, only ribosomes.



The Protective Shell: The Prokaryotic Cell Envelope

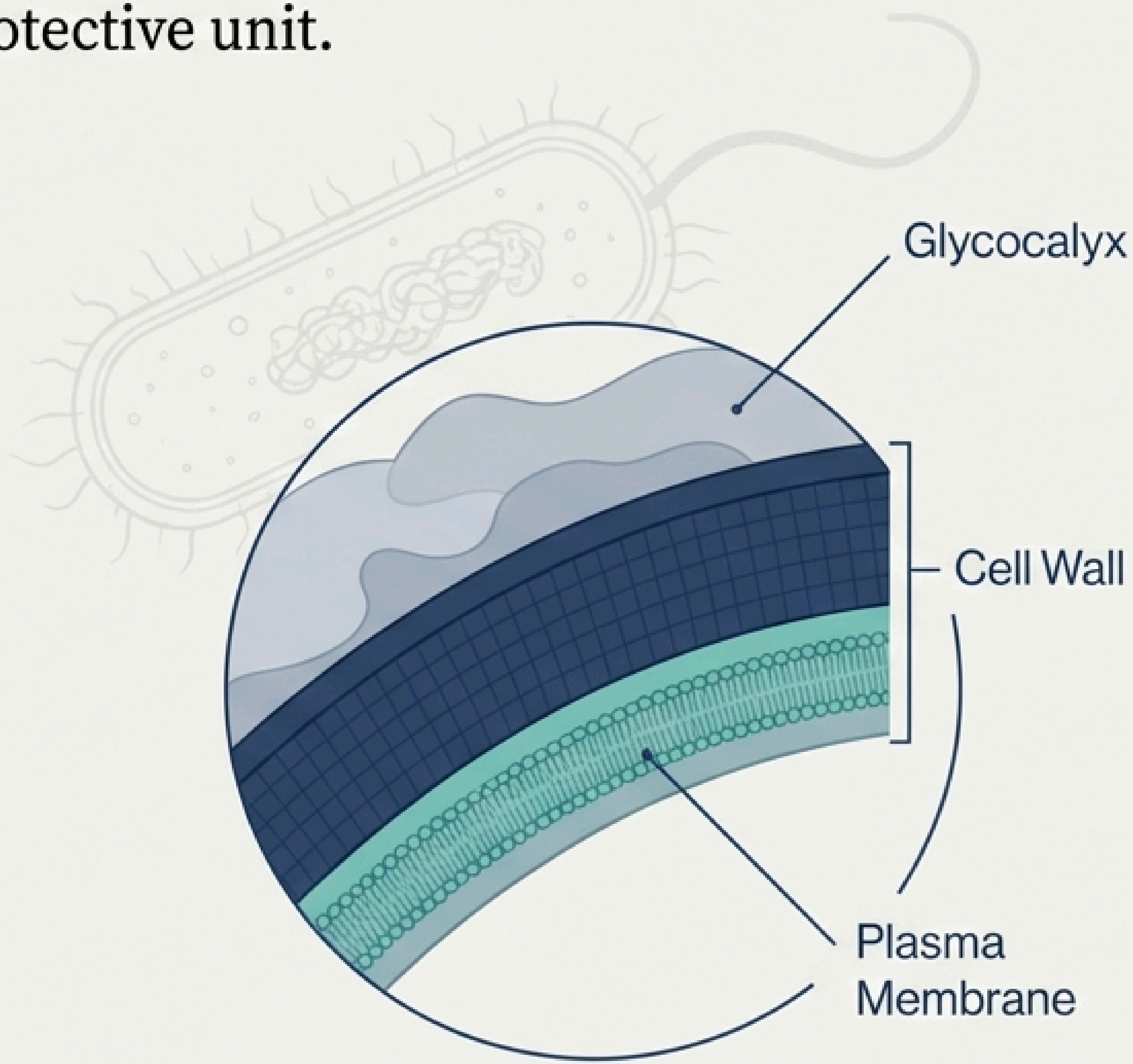
A chemically complex, tightly bound, three-layered structure that acts as a single protective unit.

The Layers (from outside in):

1. Outermost: Glycocalyx
- Differs in composition and thickness.
- **Slime Layer:** A loose sheath.
- **Capsule:** A thick and tough layer.

2. Middle: Cell Wall
- **Function:** Determines the cell's shape and provides strong structural support to prevent bursting or collapsing.
- **Key Exception: Mycoplasma** is an organism that lacks a cell wall.

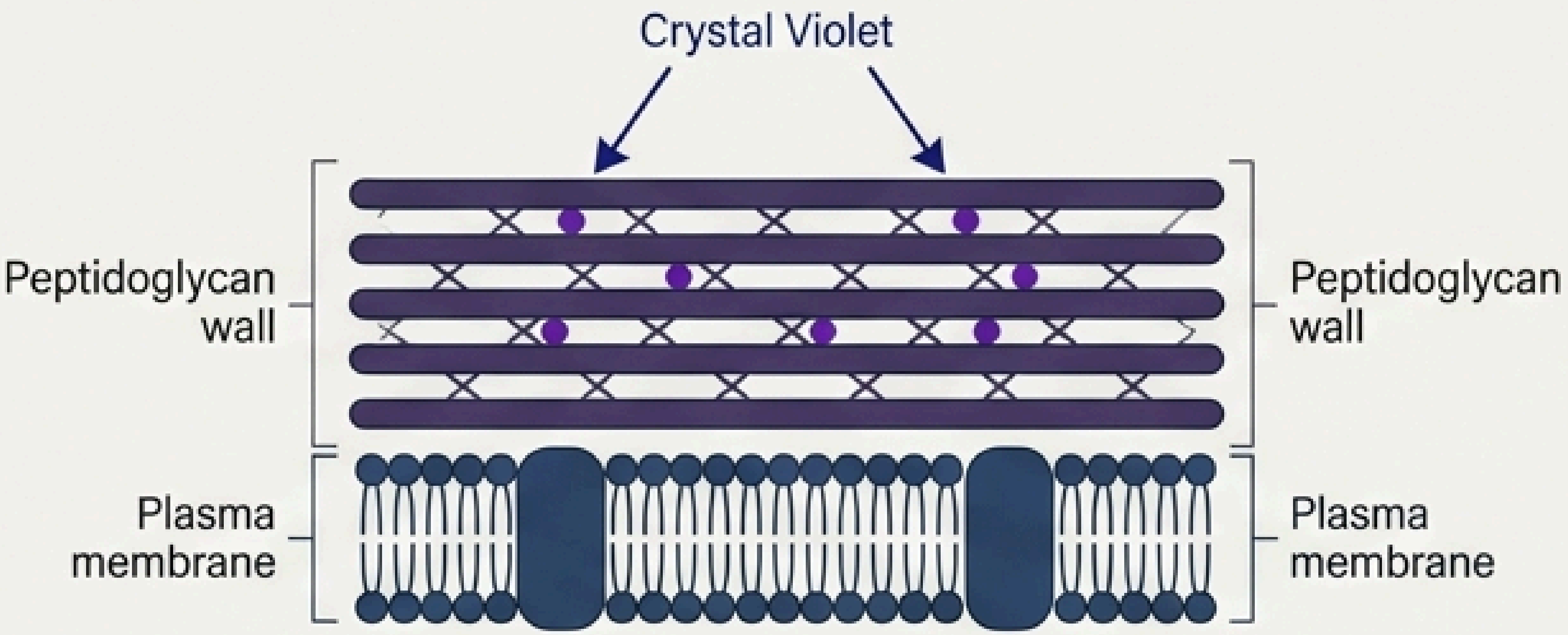
3. Innermost: Plasma Membrane
- **Nature:** Selectively permeable, interacting with the outside world.
- **Structure:** Structurally similar to the eukaryotic membrane.



The Gram Stain: Differentiating by the Wall

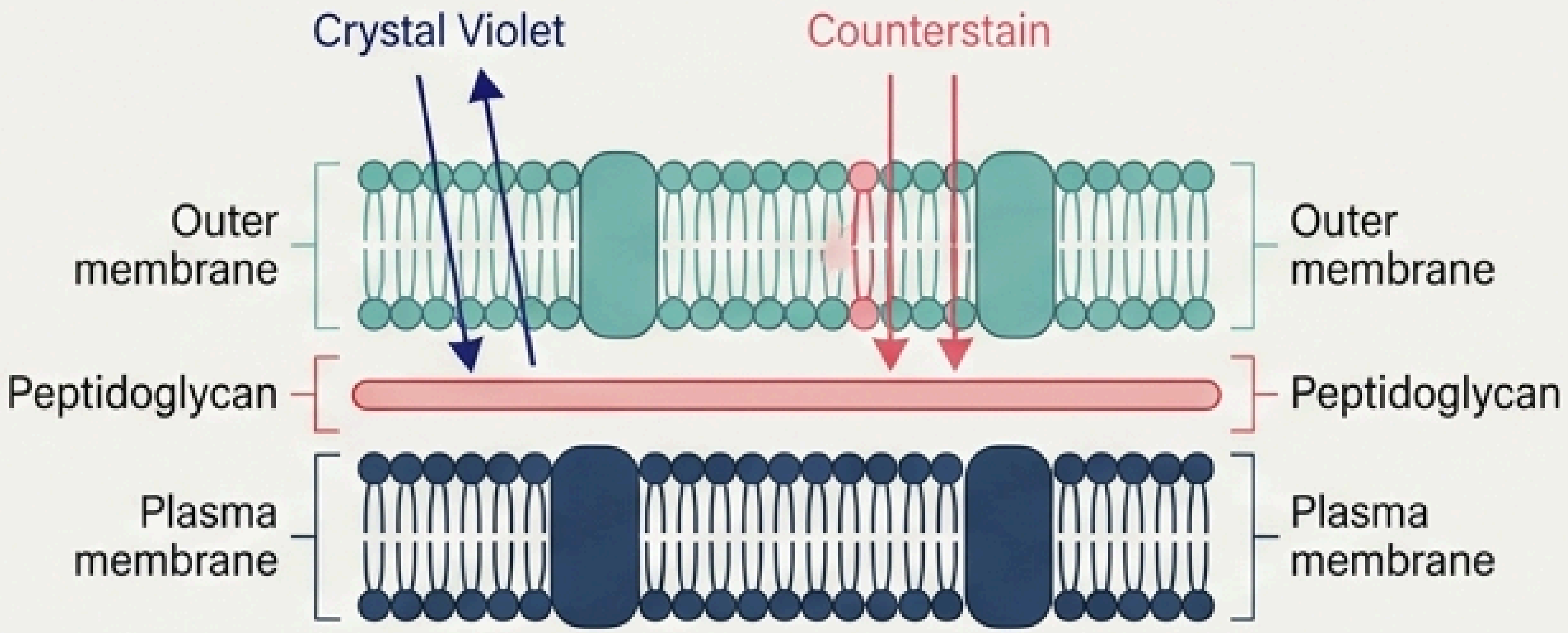
Bacteria are classified into two groups based on how their cell envelopes respond to the Gram staining procedure.

Gram-Positive Bacteria



- Take up and retain the gram stain (crystal violet).
- Appear purple.
- The reason for retention is the difference in their cell envelope structure.

Gram-Negative Bacteria



- Do not retain the gram stain.
- Appear pink/red after counterstaining.
- The reason for not retaining is the difference in their cell envelope structure.

Key Takeaway: The classification as Gram-positive or Gram-negative is a direct result of structural differences in their cell envelopes.

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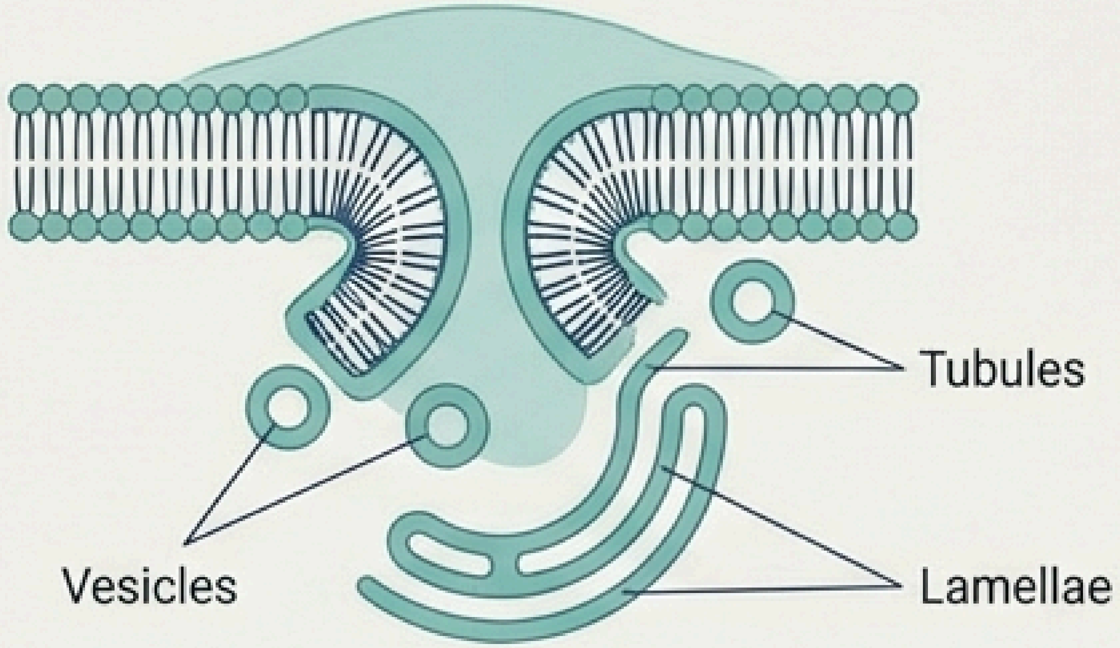
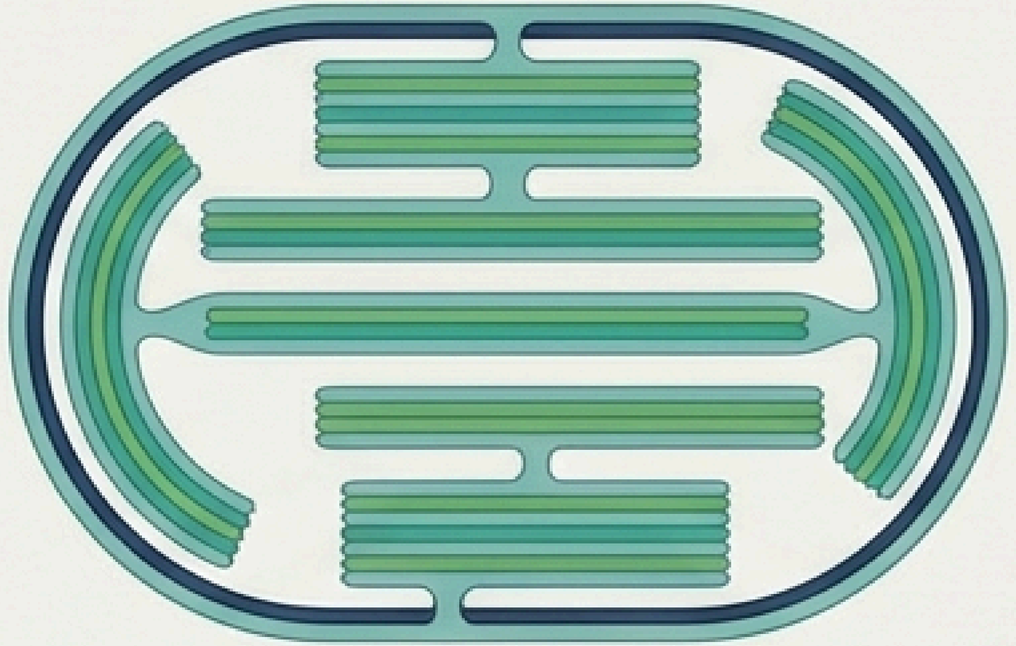
Smart Folds: Expanding Membrane Function



1. Mesosomes

- **Formation:** A special membranous structure formed by the extensions of the plasma membrane into the cell.
- **Forms:** Vesicles, tubules, and lamellae.
- **Crucial Functions:**
  - Cell wall formation.
  - DNA replication and distribution to daughter cells.
  - Respiration and secretion processes.
  - Increases the surface area of the plasma membrane and enzymatic content.



2. Chromatophores

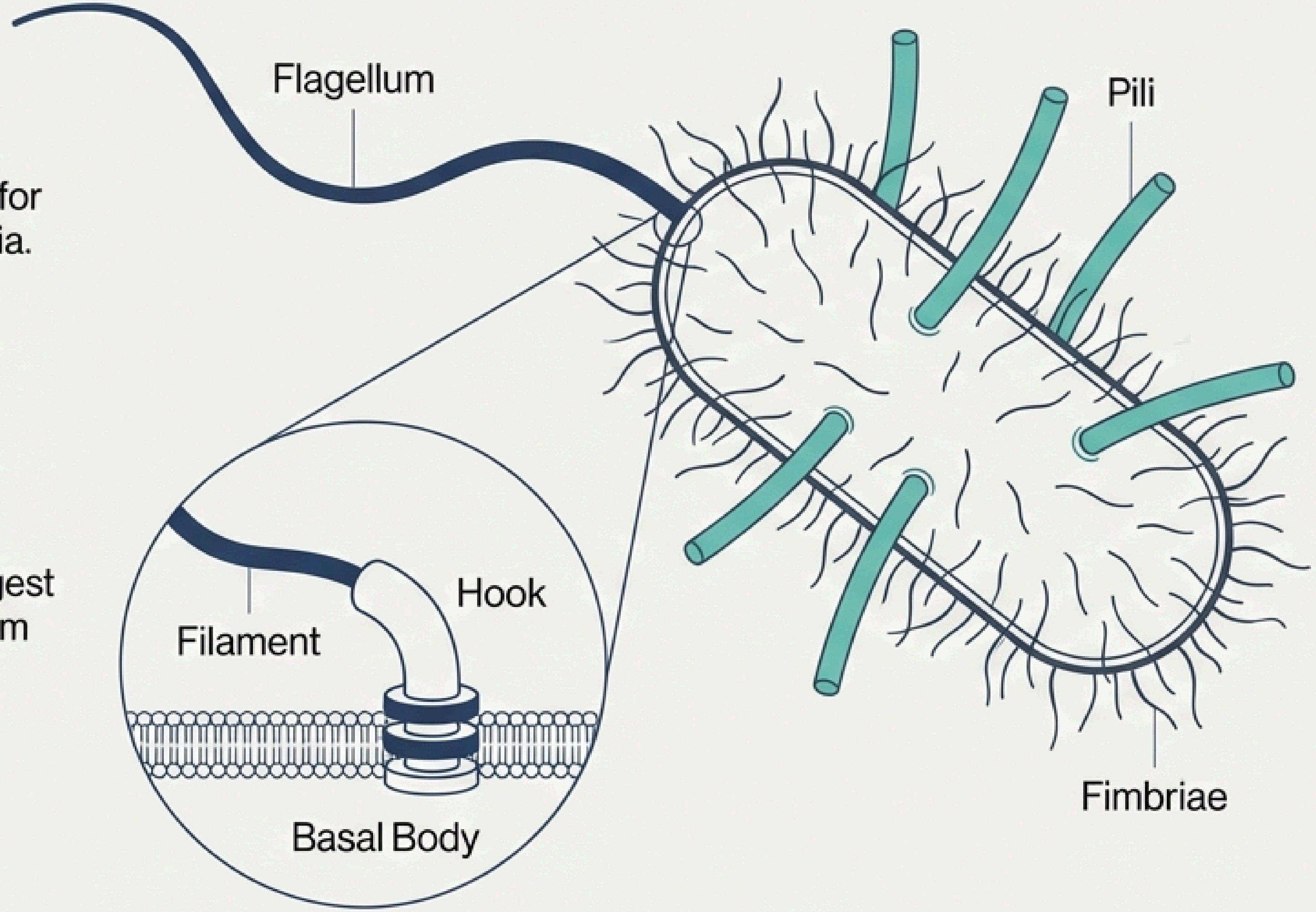
- **Found In:** Some prokaryotes like cyanobacteria.
- **Structure:** Other membranous extensions into the cytoplasm.
- **Function:** Contain pigments.

Surface Structures: Tools for Interaction

For Motility

Flagella

- **Function:** Responsible for motility in motile bacteria.
- **Structure:** Thin, filamentous extensions from the cell wall.
- **Three Parts:** Filament, Hook, and Basal Body.
- The Filament is the longest portion and extends from the cell surface to the outside.



For Attachment (Not Motility)

Pili

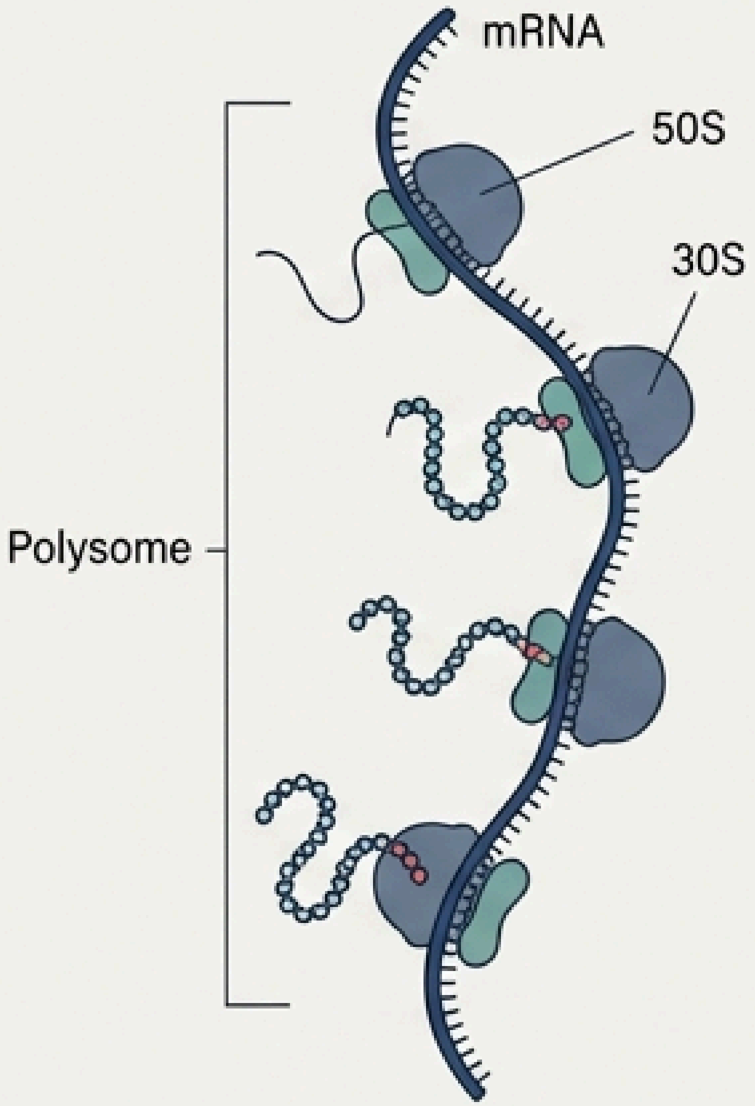
- **Structure:** Elongated tubular structures made of a special protein.

Fimbriae

- **Structure:** Small, bristle-like fibers sprouting out of the cell.
- **Function:** Help attach bacteria to surfaces like rocks in streams and also to host tissues.

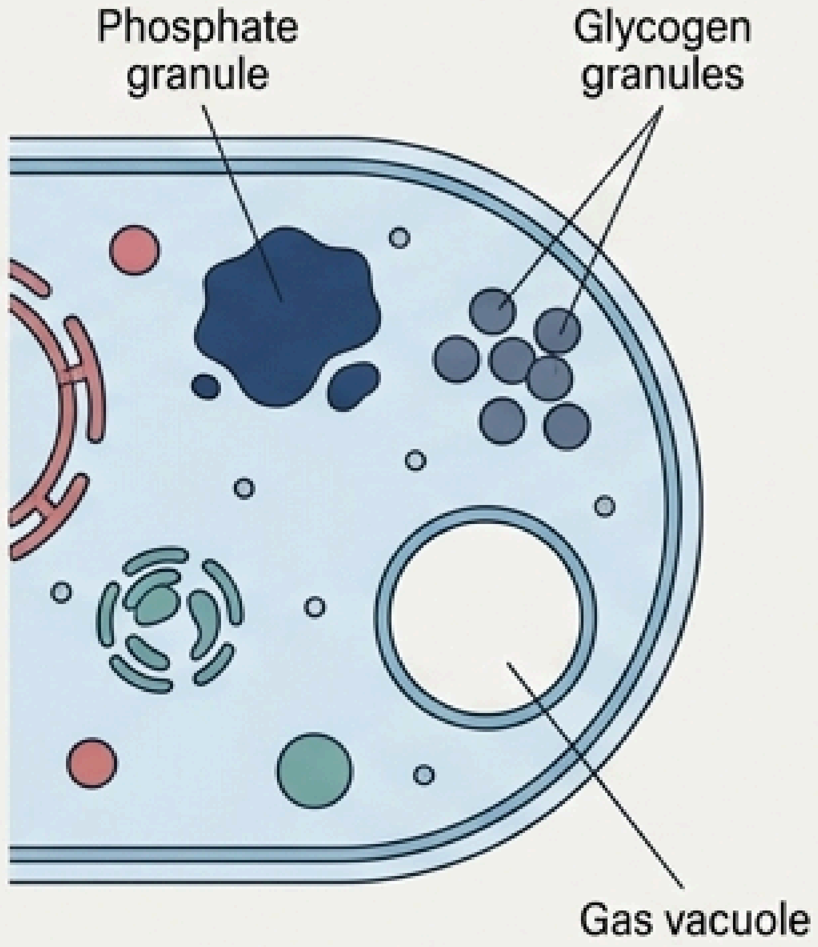
Cytoplasmic Contents: Factories & Storage

Ribosomes



- **Location:** Associated with the plasma membrane.
- **Size:** About 15 nm by 20 nm.
- **Function:** The site of protein synthesis.
- **Structure:**
  - Prokaryotic ribosomes are **70S** type.
  - Made of two subunits: **50S** (large) and **30S** (small).
- **Polysomes (or Polyribosomes):** A chain of several ribosomes attached to a single mRNA, translating it into proteins.

Inclusion Bodies



- **Function:** Store reserve materials in the cytoplasm.
- **Key Feature:** Are not bound by any membrane system and lie free in the cytoplasm.
- **Examples:** Phosphate granules, cyanophycean granules, glycogen granules.
- **Gas Vacuoles:** Found in blue-green and purple/green photosynthetic bacteria.

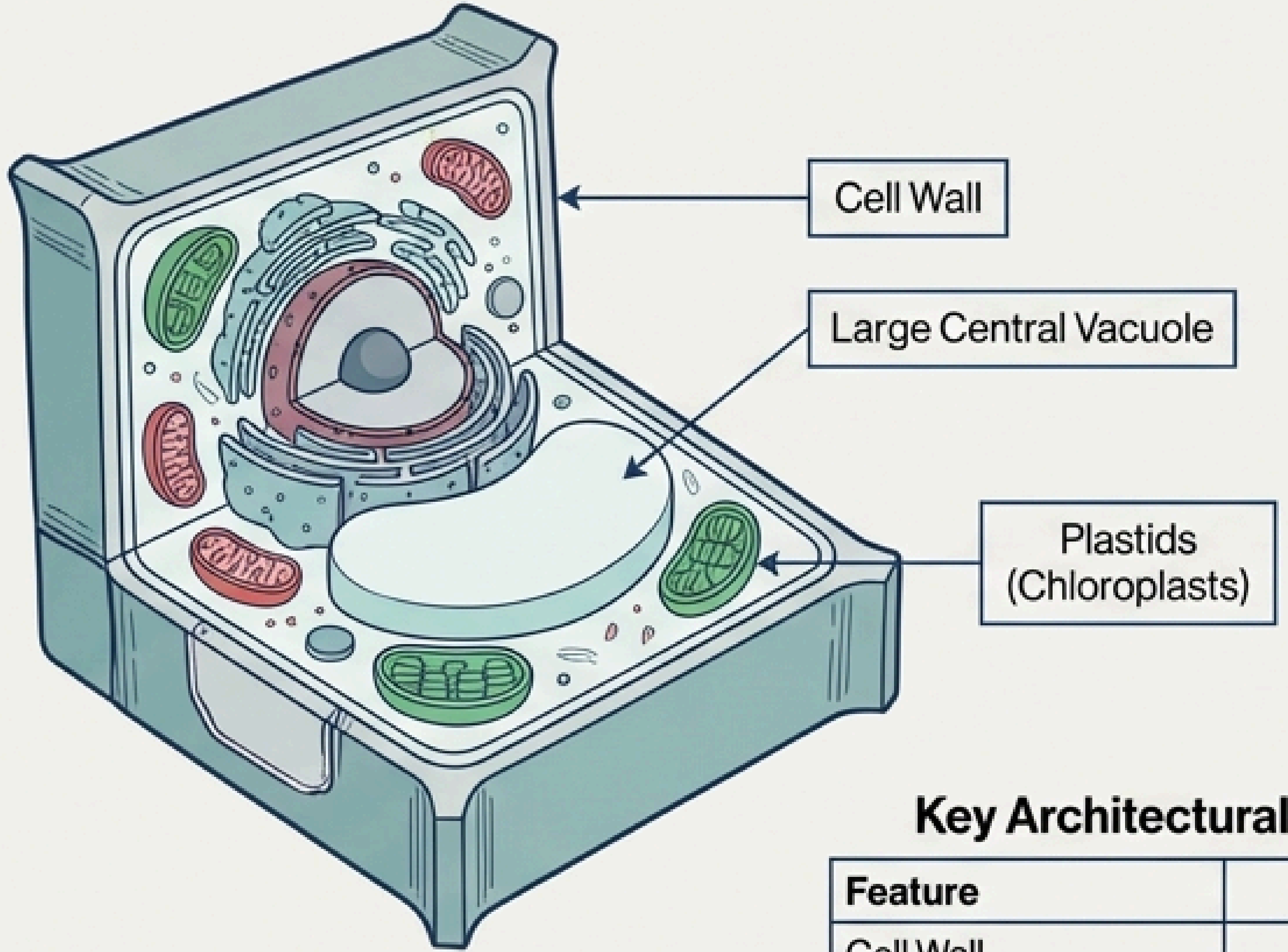
The Eukaryotic Leap: An Organized Metropolis

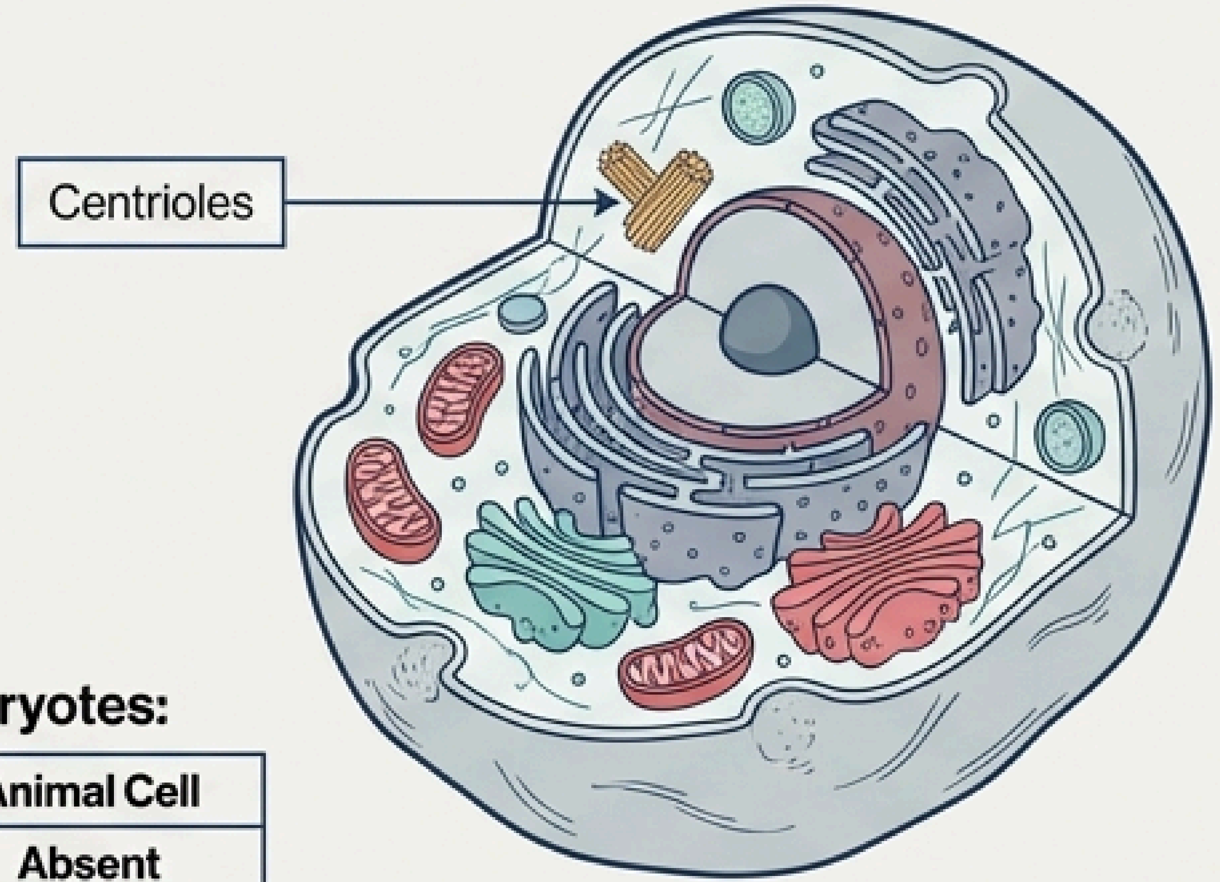
Inhabitants

Protists, Fungi, Plants, and Animals.

Defining Features

- **Organized Nucleus:** An organized nucleus with a nuclear envelope. Genetic material is organized into chromosomes.
- **Compartmentalization:** Extensive presence of membrane-bound organelles.
- **Complex Structures:** A variety of complex locomotory and cytoskeletal structures.





Key Architectural Differences Within Eukaryotes:

Feature	Plant Cell	Animal Cell
Cell Wall	**Present**	**Absent**
Plastids	**Present**	**Absent**
Large Central Vacuole	**Present**	**Absent**
Centrioles	**Absent** in almost all plant cells	**Present**

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The Fluid Mosaic Model: A Dynamic Barrier

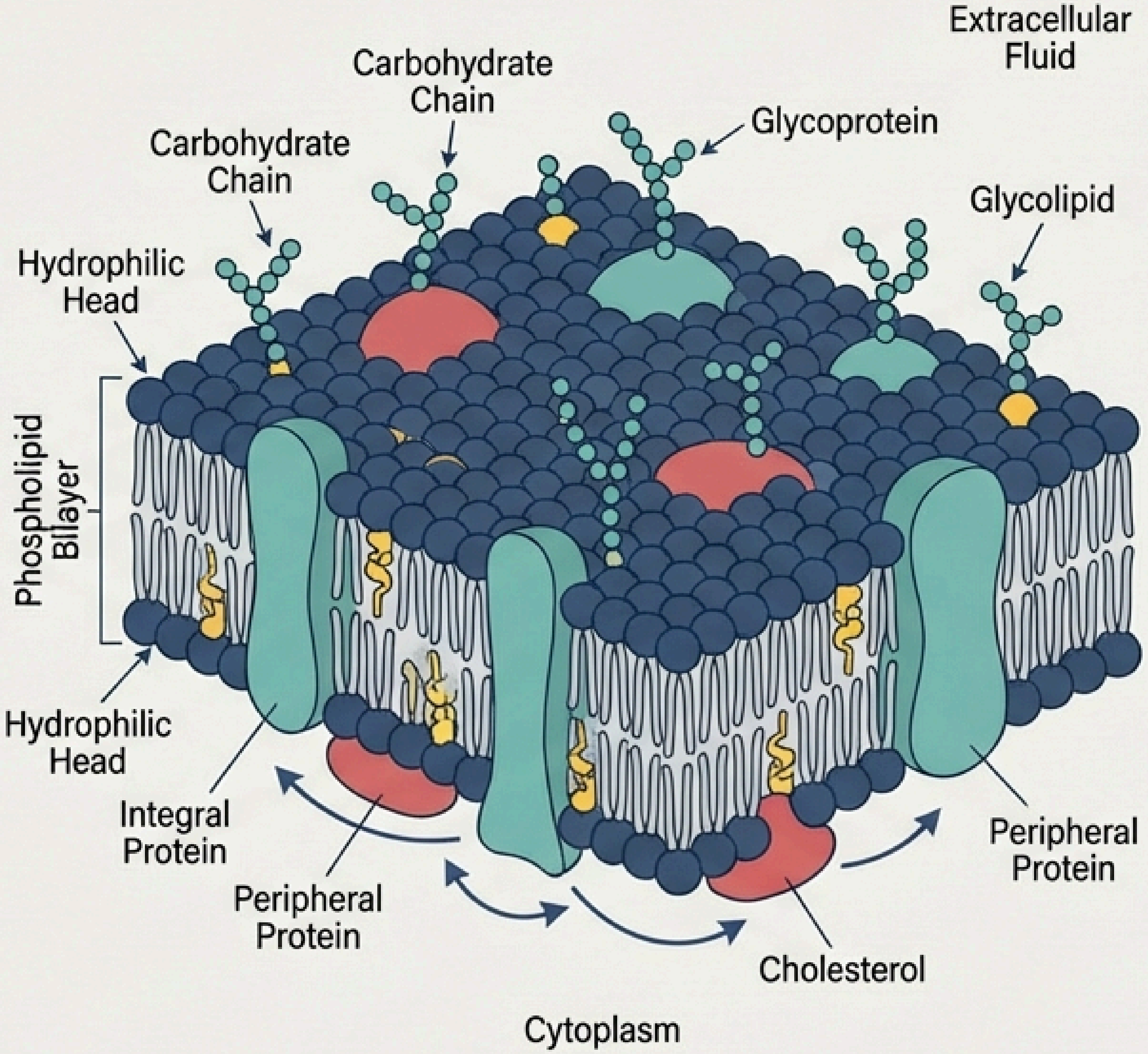
Historical Context: Proposed by Singer and Nicolson (1972) after studies on human red blood cells (RBCs) in the 1950s.

Chemical Composition:

- **Lipids:** Major lipids are phospholipids arranged in a bilayer.
- **Arrangement:** The polar (hydrophilic) head faces outwards; the hydrophobic (nonpolar) tails face the inner part, protected from the aqueous environment. Also contains cholesterol.
- **Proteins:**
  - **Peripheral:** Lie on the surface.
  - **Integral:** Partially or totally buried in the membrane.
- **Ratio Varies:** In human RBCs, the membrane is approximately 52% protein and 40% lipid.

The "Fluid" Nature:

- The quasi-fluid nature of lipid enables lateral movement of proteins within the bilayer.
- **Functional Importance:** This fluidity is crucial for cell growth, formation of intercellular junctions, secretion, and cell division.



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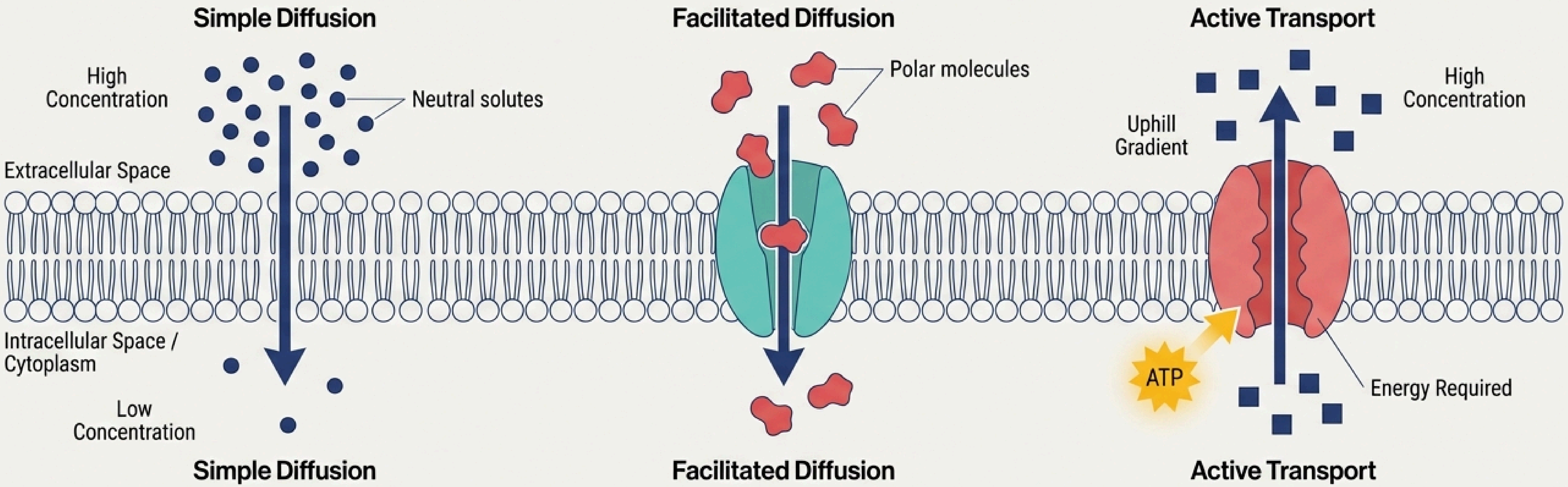
Crossing the Border: Membrane Transport

The plasma membrane is **selectively permeable.**



1. Passive Transport (No Energy Required)

- Movement along the concentration gradient (High → Low).
- **Simple Diffusion:** Movement of neutral solutes across the membrane.
- Osmosis: The specific term for the movement of water by diffusion.
- **Facilitated Diffusion:** Polar molecules cannot pass through the nonpolar lipid bilayer, so they require a carrier protein to facilitate transport.

2. Active Transport (Requires Energy - ATP)

- Movement against the concentration gradient (Low → High).
- An energy-dependent process where ATP is utilized.
- **Classic Example:** The Na^+/K^+ Pump.

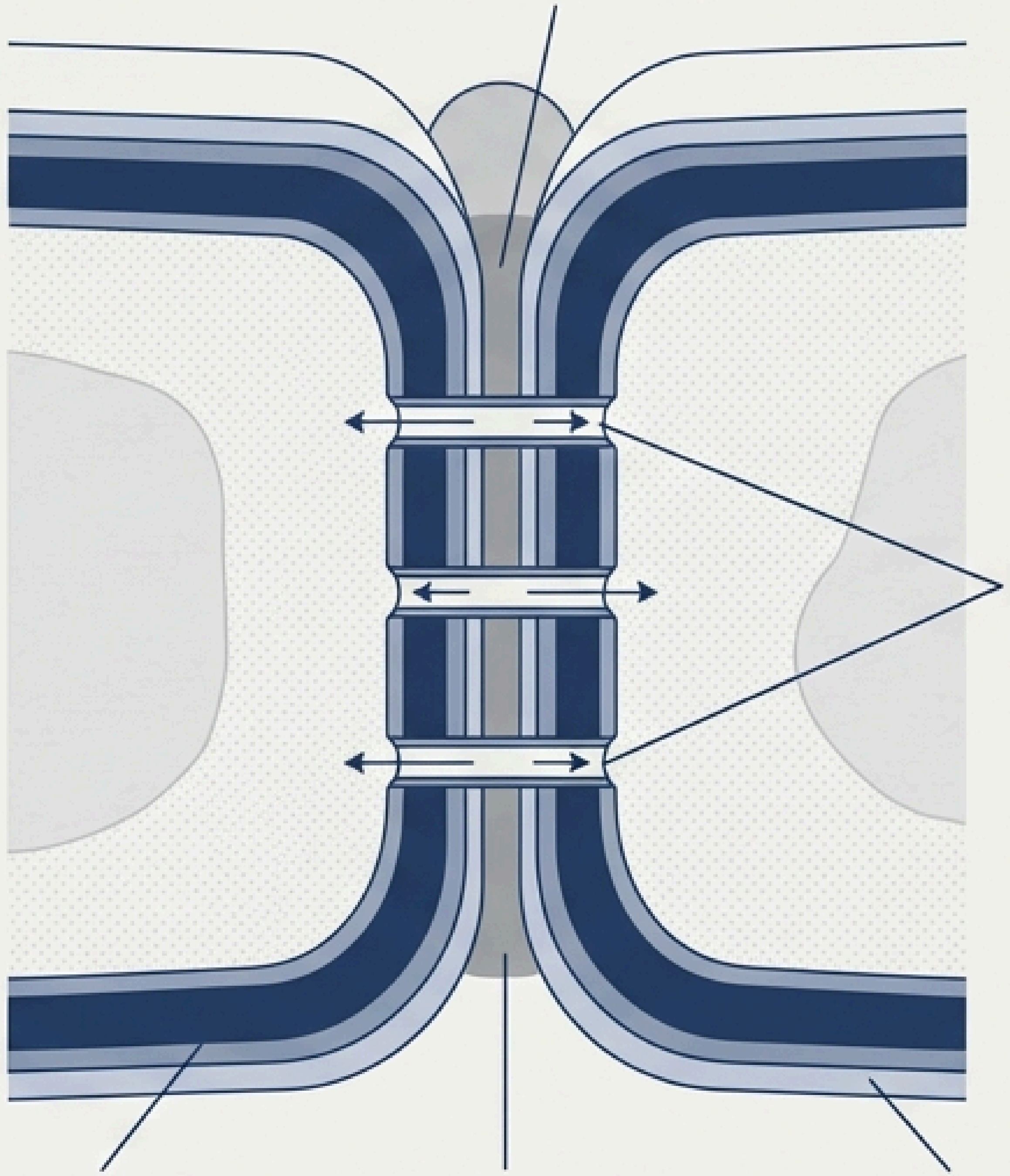
The Cell Wall: Rigid Support and Connection

Core Concept: A non-living, rigid structure that forms an outer covering for the plasma membrane of fungi and mọnns an outer covering for the plasma membrane of fungi and plants.

Functions:
Gives shape to the cell.
Protects from mechanical damage and infection.
Helps in cell-to-cell interaction.
Provides a barrier to undesirable macromolecules.

Composition Varies:
- **Algae:** Cellulose, galactans, mannans, and minerals like calcium carbonate.
- **Other Plants:** Cellulose, hemicellulose, pectins, and proteins.

Layers of the Plant Cell Wall:
- **Primary Wall:** Found in young plant cells; capable of growth.
- **Secondary Wall:** Formed on the inner (towards membrane) side as the cell matures.
- **Middle Lamella:** A layer mainly of **calcium pectate** which holds or glues the different neighbouring cells together.
- **Plasmodesmata:** Connect the cytoplasm of neighbouring cells.



Key Architectural Differences: A Summary for Revision

Feature	Prokaryote	Eukaryote
Genetic Material	• Nucleoid (no envelope), single circular DNA, plasmids	• True nucleus (with envelope), multiple chromosomes
Cell Wall	• Complex (peptidoglycan), present in most	• Simple (cellulose/chitin), present in plants & fungi
Ribosomes	• 70S (50S + 30S)	• 80S in cytoplasm (60S + 40S), 70S in organelles
Membrane Infoldings	• Mesosomes (respiration, replication)	• Not present (functions handled by organelles)
Organelles	• Absent	• Present (extensive compartmentalization)
Reserve Material	• Non-membranous Inclusion Bodies	• Vacuoles, granules
Attachment	• Pili and Fimbriae	• Intercellular junctions

NEET-UG Practice MCQs

1. The naked, single circular chromosome is a characteristic of which cell type?
   a) Plant Cells b) Animal Cells c) Eukaryotic Cells d) Prokaryotic Cells
2. Which structure in bacteria confers unique phenotypic characters like antibiotic resistance?
   a) Nucleoid b) Mesosome c) Plasmid d) Ribosome
3. A thick and tough glycocalyx layer in bacteria is known as the:
   a) Slime layer b) Capsule c) Cell wall d) Plasma membrane
4. The specialized infoldings of the plasma membrane in prokaryotes that help in DNA replication are called:
   a) Chromatophores b) Plasmids c) Fimbriae d) Mesosomes
5. Prokaryotic ribosomes are of which type?
   a) 80S (60S + 40S) b) 70S (50S + 30S) c) 70S (40S + 30S) d) 80S (50S + 30S)
6. In the Fluid Mosaic Model, the quasi-fluid nature of what component allows for the lateral movement of proteins?
   a) Carbohydrates b) Peripheral proteins c) Lipids d) Integral proteins
7. Transport of a molecule against its concentration gradient, requiring ATP, is known as:
   a) Osmosis b) Active transport c) Simple diffusion d) Facilitated diffusion
8. The middle lamella, which glues neighbouring plant cells together, is primarily composed of:
   a) Cellulose b) Calcium carbonate c) Hemicellulose d) Calcium pectate
9. Which of the following is present in animal cells but absent in almost all plant cells?
   a) Plastids b) Large central vacuole c) Centrioles d) Cell wall
10. Gas vacuroles, which are not bound by any membrane, are found in:
    a) All bacteria b) Human Red Blood Cells c) Blue-green photosynthetic bacteria d) All eukaryotic cells

Answer Key

1. **d)** Prokaryotic Cells
2. **c)** Plasmid
3. **b)** Capsule
4. **d)** Mesosomes
5. **b)** 70S (50S + 30S)
6. **c)** Lipids
7. **b)** Active transport
8. **d)** Calcium pectate
9. **c)** Centrioles
10. **c)** Blue-green photosynthetic bacteria

All the Best for Your Preparation!